SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Class Test

Duration: 1hr

1. A warehouse robot must determine the shortest path to transport goods while avoiding obstacles. Apply **Dynamic Programming** to develop an optimal policy and explain how value iteration can improve navigation efficiency**.** **(5 Marks)**

2. A recommendation system collects customer interactions over multiple sessions. Apply **Monte Carlo Control** to update the policy and explain how the agent improves recommendations with experience. **.** **(5 Marks)**

3. A delivery robot operates in an uncertain environment. Apply the **SARSA** algorithm to update the action-value function and explain how the learned policy adapts to environmental changes. **.**
(5 Marks)

4. A robotic arm performs continuous object manipulation. Apply the **REINFORCE** algorithm to develop a policy that maximizes task completion accuracy. **.**

(5 Marks)

5. An autonomous vehicle is trained using **Q-Learning** and **Deep Q-Networks (DQN)**. Analyze the differences in learning performance, convergence speed, and scalability for complex driving environments. **.** **(5 Marks)**

6. A game-playing AI is implemented using **DQN**, **Double DQN (DDQN)**, and **Dueling DQN**. Analyze the advantages of DDQN and Dueling DQN over the standard DQN architecture. **.** **(5 Marks)**

7. A smart drone navigation system uses **A2C** and **A3C** algorithms. Analyze the impact of parallel learning on convergence speed, computational efficiency, and policy stability. **.** **(5 Marks)**

8. A healthcare decision-support system operates with incomplete patient information. Analyze how a Partially Observable Markov Decision Process (POMDP) improves decision-making compared to a standard MDP. **.** **(5 Marks)**

9. A robotic navigation system is trained using PPO, TRPO, and DDPG. Evaluate the suitability of each algorithm for continuous control tasks based on stability, sample efficiency, and computational complexity. **.** **(5 Marks)**

10. A smart city traffic management system uses **Multi-Agent Reinforcement Learning** (MARL) to coordinate traffic signals. Evaluate the **effectiveness of MARL** in improving traffic flow while considering scalability, coordination, fairness, and ethical challenges. **.** **(5 Marks)**

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively